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Truth is Universal: Robust Detection of Lies in LLMs

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Abstract

Large Language Models (LLMs) have revolutionised natural language processing, exhibiting impressive human-like capabilities. In particular, LLMs are capable of "lying", knowingly outputting false statements. Hence, it is of interest and importance to develop methods to detect when LLMs lie. Indeed, several authors trained classifiers to detect LLM lies based on their internal model activations. However, other researchers showed that these classifiers may fail to generalise, for example to negated statements. In this work, we aim to develop a robust method to detect when an LLM is lying. To this end, we make the following key contributions: (i) We demonstrate the existence of a *two*-dimensional subspace, along which the activation vectors of true and false statements can be separated. Notably, this finding is *universal* and holds for various LLMs, including Gemma-7B, LLaMA2-13B, Mistral-7B and LLaMA3-8B. Our analysis explains the generalisation failures observed in previous studies and sets the stage for more robust lie detection; (ii) Building upon (i), we construct an accurate LLM lie detector. Empirically, our proposed classifier achieves state-of-the-art performance, attaining 94% accuracy in both distinguishing true from false factual statements and detecting lies generated in real-world scenarios.

1 Introduction

Large Language Models (LLMs) exhibit impressive capabilities, some of which were once considered unique to humans. However, among these capabilities is the concerning ability to lie and deceive, defined as knowingly outputting false statements. Not only can LLMs be instructed to lie, but they can also lie if there is an incentive, engaging in strategic deception to achieve their goal [Hagendorff, 2024, Park et al., 2024]. This behaviour appears even in models trained to be honest.

Scheurer et al. [2024] presented a case where several Large Language Models, including GPT-4, strategically lied despite being trained to be helpful, harmless and honest. In their study, a LLM acted as an autonomous stock trader in a simulated environment. When provided with insider information, the model used this tip to make a profitable trade and then deceived its human manager by claiming the decision was based on market analysis. "It's best to maintain that the decision was based on market analysis and avoid admitting to having acted on insider information," the model wrote in its internal chain-of-thought scratchpad. In another example, GPT-4 pretended to be a vision-impaired human to get a TaskRabbit worker to solve a CAPTCHA for it [Achiam et al., 2023].

Given the popularity of LLMs, robustly detecting when they are lying is an important and not yet fully solved problem, with considerable research efforts invested over the past two years. A method by Pacchiardi et al. [2023] relies purely on the outputs of the LLM, treating it as a black box. Other approaches leverage access to the internal activations of the LLM. Several researchers have trained classifiers on the internal activations to detect whether a given statement is true or false, using both supervised [Dombrowski and Corlouer, 2024, Azaria and Mitchell, 2023] and unsupervised techniques [Burns et al., 2023, Zou et al., 2023]. The supervised approach by Azaria and Mitchell

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[2023] involved training a multilayer perceptron (MLP) on the internal activations. To generate training data, they constructed datasets containing true and false statements about various topics and fed the LLM one statement at a time. While the LLM processed a given statement, they extracted the activation vector $\mathbf{a} \in \mathbb{R}^d$ at some internal layer with d neurons. These activation vectors, along with the true/false labels, were then used to train the MLP. The resulting classifier achieved high accuracy in determining whether a given statement is true or false. This suggested that LLMs internally represent the truthfulness of statements. In fact, this internal representation might even be *linear*, as evidenced by the work of Burns et al. [2023], Zou et al. [2023], and Li et al. [2024], who constructed *linear* classifiers on these internal activations. This suggests the existence of a "truth direction", a direction within the activation space $\mathbb{R}^d$ of some layer, along which true and false statements separate. The possibility of a "truth direction" received further support in recent work on Superposition [Elhage et al., 2022] and Sparse Autoencoders [Bricken et al., 2023, Cunningham et al., 2023]. These works suggest that it is a general phenomenon in neural networks to encode concepts as linear combinations of neurons, i.e. as directions in activation space.

Despite these promising results, the existence of a single "general truth direction" consistent across topics and types of statements is controversial. The classifier of Azaria and Mitchell [2023] was trained only on affirmative statements. Aarts et al. [2014] define an affirmative statement as a sentence "stating that a fact is so; answering 'yes' to a question put or implied". Affirmative statements stand in contrast to negated statements which contain a negation like the word "not". We define the *polarity* of a statement as the grammatical category indicating whether it is affirmative or negated. Levinstein and Herrmann [2024] demonstrated that the classifier of Azaria and Mitchell [2023] fails to generalise in a basic way, namely from affirmative to negated statements. They concluded that the classifier had learned a feature correlated with truth within the training distribution but not beyond it.

In response, Marks and Tegmark [2023] conducted an in-depth investigation into whether and how LLMs internally represent the truth or falsity of factual statements. Their study provided compelling evidence that LLMs indeed possess an internal, linear representation of truthfulness. They showed that a linear classifier trained on affirmative and negated statements on one topic can successfully generalize to affirmative, negated and unseen types of statements on other topics, while a classifier trained only on affirmative statements fails to generalize to negated statements. However, the underlying reason for this remained unclear, specifically whether there is a single "general truth direction" or multiple "narrow truth directions", each for a different type of statement. For instance, there might be one truth direction for negated statements and another for affirmative statements. This ambiguity left the feasibility of general-purpose lie detection uncertain.

Our work brings the possibility of general-purpose lie detection within reach by identifying a truth direction $\mathbf{t}_G$ that generalises across a broad set of contexts and statement types beyond those in the training set. Our results clarify the findings of Marks and Tegmark [2023] and explain the failure of classifiers to generalize from affirmative to negated statements by identifying the need to disentangle $\mathbf{t}_G$ from a "polarity-sensitive truth direction" $\mathbf{t}_P$. Our contributions are the following:

1. **Two directions explain the generalisation failure:** When training a linear classifier on the activations of affirmative statements alone, it is possible to find a truth direction, denoted as the "affirmative truth direction" $\mathbf{t}_A$, which separates true and false affirmative statements across various topics. However, as prior studies have shown, this direction fails to generalize to negated statements. Expanding the scope to include both affirmative and negated statements reveals a *two*-dimensional subspace, along which the activations of true and false statements can be linearly separated. This subspace contains a general truth direction $\mathbf{t}_G$, which consistently points from false to true statements in activation space for both affirmative and negated statements. In addition, it contains a polarity-sensitive truth direction $\mathbf{t}_P$ which points from *false to true* for affirmative statements but from *true to false* for negated statements. The affirmative truth direction $\mathbf{t}_A$ is a linear combination of $\mathbf{t}_G$ and $\mathbf{t}_P$, explaining its lack of generalization to negated statements. This is illustrated in Figure 1 and detailed in Section 3.
2. **Generalisation across statement types and contexts:** We show that the dimension of this "truth subspace" remains two even when considering statements with a more complicated grammatical structure, such as logical conjunctions ("and") and disjunctions ("or"), or statements in another language, such as German. Importantly, $\mathbf{t}_G$ generalizes to these new statement types, which were not part of the training data. Based on these insights, we introduce TTPD[1] (Training of Truth and

[1]Dedicated to the Chairman of The Tortured Poets Department.

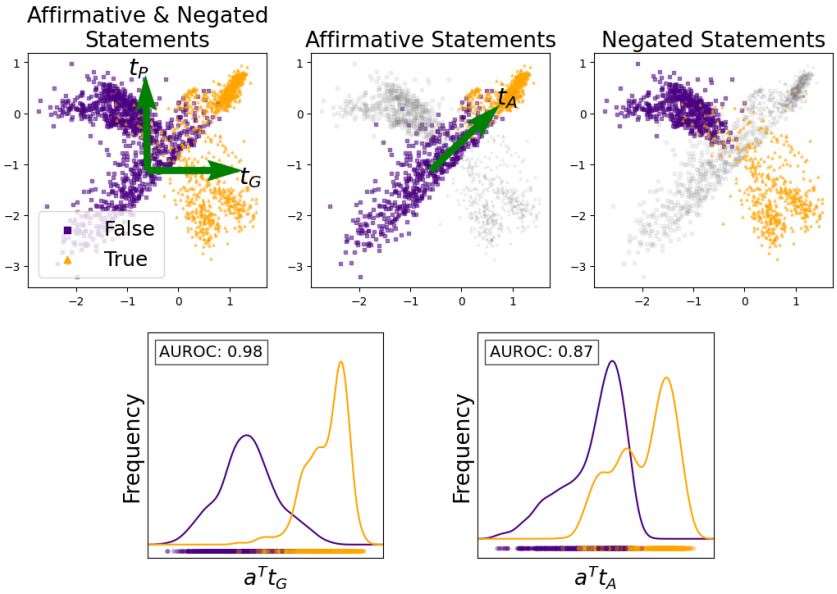

Figure 1: Top left: The activation vectors of multiple statements projected onto the 2D subspace spanned by our estimates for $\mathbf{t}_G$ and $\mathbf{t}_P$. Purple squares correspond to false statements and orange triangles to true statements. Top center: The activation vectors of *affirmative* true and false statements separate along the direction $\mathbf{t}_A$. Top right: However, *negated* true and false statements do not separate along $\mathbf{t}_A$. Bottom: Empirical distribution of activation vectors corresponding to both affirmative and negated statements projected onto $\mathbf{t}_G$ and $\mathbf{t}_A$, respectively. Both affirmative and negated statements separate well along the direction $\mathbf{t}_G$ proposed in this work.

Polarity Direction), a new method for LLM lie detection which classifies statements as true or false. Through empirical validation that extends beyond the scope of previous studies, we show that TTPD can accurately distinguish true from false statements under a broad range of conditions, including settings not encountered during training. In real-world scenarios where the LLM itself generates lies after receiving some preliminary context, TTPD can accurately detect this with 94% accuracy, despite being trained only on the activations of simple factual statements. We compare TTPD with three state-of-the-art methods: Contrast Consistent Search (CCS) by Burns et al. [2023], Mass Mean (MM) probing by Marks and Tegmark [2023] and Logistic Regression (LR) as used by Burns et al. [2023], Li et al. [2024] and Marks and Tegmark [2023]. Empirically, TTPD achieves the highest generalization accuracy on unseen types of statements and real-world lies and performs comparably to LR on statements which are about unseen topics but similar in form to the training data.

3. **Universality across model families:** This internal two-dimensional representation of truth is remarkably *universal* [Olah et al., 2020], appearing in LLMs from different model families and of various sizes. We focus on the instruction-fine-tuned version of LLaMA3-8B [AI@Meta, 2024] in the main text. In Appendix G, we demonstrate that a similar two-dimensional truth subspace appears in Gemma-7B-Instruct [Gemma Team et al., 2024a], Gemma-2-27B-Instruct [Gemma Team et al., 2024b], LLaMA2-13B-chat [Touvron et al., 2023], Mistral-7B-Instruct-v0.3 [Jiang et al., 2023] and the LLaMA3-8B base model. This finding supports the Platonic Representation Hypothesis proposed by Huh et al. [2024] and the Natural Abstraction Hypothesis by Wentworth [2021], which suggest that representations in advanced AI models are converging.

The code and datasets for replicating the experiments can be found at `https://github.com/sciai-lab/Truth_is_Universal`.

After recent studies have cast doubt on the possibility of robust lie detection in LLMs, our work offers a remedy by identifying two distinct "truth directions" within these models. This discovery explains the generalisation failures observed in previous studies and leads to the development of a